



CHE 109: Engineering Chemistry – I
Summer-2025
Electrochemistry
Chapter 7

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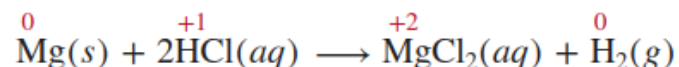
One form of energy that has tremendous practical significance is electrical energy. A day without electricity from either the power company or batteries is unimaginable in our technological society. The area of chemistry that deals with the interconversion of electrical energy and chemical energy is electrochemistry.

Electrochemical processes are redox reactions in which the energy released by a spontaneous reaction is converted to electricity or in which electricity is used to drive a nonspontaneous chemical reaction. The latter type is called electrolysis.

18.1 Redox Reactions

Electrochemistry is the branch of chemistry that deals with the interconversion of electrical energy and chemical energy. Electrochemical processes are redox (oxidation-reduction) reactions in which the energy released by a spontaneous reaction is converted to electricity or in which electrical energy is used to cause a nonspontaneous reaction to occur. Although redox reactions were discussed in Chapter 4, it is helpful to review some of the basic concepts that will come up again in this chapter.

In redox reactions, electrons are transferred from one substance to another. The reaction between magnesium metal and hydrochloric acid is an example of a redox reaction:



Recall that the numbers above the elements are the oxidation numbers of the elements. The rules for assigning oxidation numbers are presented in Section 4.4. The loss of electrons by an element during oxidation is marked by an increase in the element's oxidation number. In reduction, there is a decrease in oxidation number resulting from a gain of electrons by an element. In the preceding reaction, Mg metal is oxidized and H^+ ions are reduced; the Cl^- ions are spectator ions.

► Half-Reactions

A voltaic cell employs a spontaneous oxidation–reduction reaction as a source of energy. It is constructed to separate the reaction physically into two half-reactions, one involving oxidation and the other reduction, with electrons moving from the oxidation half-reaction through the external circuit to the reduction half-reaction. ►

19.1 Balancing Oxidation–Reduction Reactions in Acidic and Basic Solutions

In Chapter 4 (Section 4.6), we introduced the *half-reaction method* of balancing simple oxidation–reduction reactions. We now extend this method to reactions that occur in acidic or basic solutions. The steps used to balance these equations successfully are built upon those presented in Chapter 4. Keep in mind that oxidation–reduction reactions involve a transfer of electrons from one species to another. For example, in the reaction described in the chapter opener, zinc metal becomes zinc(II) ion: each zinc atom loses two electrons, and each copper(II) ion becomes copper metal (copper ion gains two electrons).

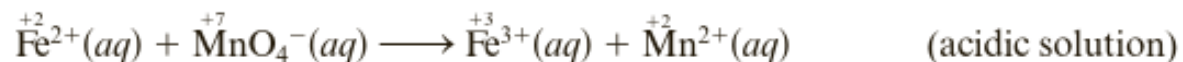
In this chapter, we will represent the hydronium ion, $\text{H}_3\text{O}^+(\text{aq})$, by its simpler notation $\text{H}^+(\text{aq})$, which chemists call the hydrogen ion. In the chapters that focused on acid–base reactions, we stressed proton transfer in solution, so we adopted the hydronium-ion notation. Here, we shift our focus to the electron-transfer aspect of certain reactions, so we adopt the simpler hydrogen-ion notation. Remember, however, that this is simply a notation change. It does not change the chemistry.

Skeleton Oxidation–Reduction Equations

To tackle more complex oxidation–reduction reactions in acidic and basic solutions, we need to review and discuss the essential information required to describe an oxidation–reduction reaction, which is called a *skeleton equation*. To set up the skeleton equation and then balance it, you need answers to the following questions:

1. What species is being oxidized (or, what is the reducing agent)? What species is being reduced (or, what is the oxidizing agent)?
2. What species result from the oxidation and reduction?
3. Does the reaction occur in acidic or basic solution?

For example, iron(II) ion may be oxidized easily by a number of oxidizing agents to yield the iron(III) ion. Permanganate ion in acidic, aqueous solution is a strong oxidizing agent and can oxidize Fe^{2+} to Fe^{3+} . In the process of oxidizing the iron(II) ion, the permanganate ion (MnO_4^-) is reduced to manganese(II) ion (Mn^{2+}). We can express these facts in the following skeleton equation:



We have written oxidation numbers over the appropriate atoms. Note that the skeleton equation is not balanced. Nor is it complete; in an acidic, aqueous solution, $\text{H}^+(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$ may be possible reactants or products in the equation. (If this were a basic solution, OH^- ion would replace the H^+ ion.)

Steps in Balancing Oxidation–Reduction Equations in Acidic Solution

Step 1: Assign oxidation numbers to each atom so you know what is oxidized and what is reduced.

Step 2: Split the skeleton equation into two half-reactions, proceeding as follows. Note the species containing the element that increases in oxidation number, and write those species to give the oxidation half-reaction. Similarly, note the species containing the element that decreases in oxidation number, and write the reduction half-reaction.

Step 3: Complete and balance each half-reaction.

- a** Balance all atoms except O and H.
- b** Balance O atoms by adding H_2O 's to one side of the equation.
- c** Balance H atoms by adding H^+ ions to one side of the equation.
- d** Balance electric charge by adding electrons (e^-) to the more positive side.

Step 4: Combine the two half-reactions to obtain the final balanced oxidation–reduction equation.

- a** Multiply each half-reaction by a factor such that when the half-reactions are added, the electrons cancel. (Electrons cannot appear in the final equation.)
- b** Simplify the balanced equation by canceling species that occur on both sides, and reduce the coefficients to the smallest whole numbers. Check that the equation is indeed balanced.

Let us see how we would apply this method to the skeleton equation we wrote for the oxidation of iron(II) ion by permanganate ion in acidic solution. We have already determined the elements that change oxidation number (Step 1). We now write the incomplete equation representing the oxidation half-reaction. Iron(II) is oxidized to iron(III) ion. Therefore, following Step 2, we write



Note that the equation is balanced in Fe as it stands (Step 3a). Steps 3b and 3c require that the half-reaction be balanced in O and H. However, since these atoms do not

appear in the half-reaction, we can ignore those steps. Then (Step 3d) you need to balance the half-reaction in electric charge. You do that by adding one electron to the more positive side.



Check that the half-reaction is balanced in Fe and in electric charge. Note that electrons are lost, as you expect for oxidation.

Now let us look at the reduction half-reaction. From the skeleton equation, you write the species containing the atom for which the oxidation number decreases (Step 2):



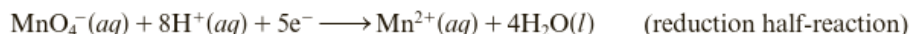
The equation is balanced in Mn as it stands (Step 3a). For Step 3b, you add four H_2O 's to the right side of the equation to balance the O atoms. (Water is present, since this is an aqueous solution.)



Step 3c says to balance the H atoms by adding H^{+} ions. (Hydrogen ions are present, since this is an acidic solution.) You add eight H^{+} ions to the left side of the equation.

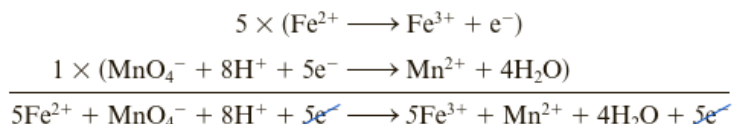


Finally, you balance the half-reaction in electric charge by adding electrons to the more positive side of the equation (Step 3d). Note that the left side has a charge of $(-1 + 8) = +7$, whereas the right side has a charge of $+2$. So you add five electrons to the left side:



Check that this half-reaction is balanced in the different kinds of atoms and in electric charge. Note that electrons are gained, as you expect for reduction.

In Step 4a, you need to multiply each half-reaction by a factor such that when the two half-reactions are added, the electrons cancel. You multiply the oxidation half-reaction by 5 (the number of electrons in the reduction half-reaction) and multiply the reduction half-reaction by 1 (the number of electrons in the oxidation half-reaction). Then you add the two half-reactions:



After canceling electrons, the equation is in simplest form (Step 4b). The final balanced equation is



You can write this equation in terms of the hydronium-ion notation, $\text{H}_3\text{O}^{+}(\text{aq})$, if you wish, by changing $8\text{H}^{+}(\text{aq})$ to $8\text{H}_3\text{O}^{+}(\text{aq})$ and adding $8\text{H}_2\text{O}(\text{l})$ to the other side of the equation to give a total of $12\text{H}_2\text{O}(\text{l})$.

18.2 Galvanic Cells

In Section 4.4 we saw that when a piece of zinc metal is placed in a CuSO_4 solution, Zn is oxidized to Zn^{2+} ions while Cu^{2+} ions are reduced to metallic copper (see Figure 4.10):



The electrons are transferred directly from the reducing agent (Zn) to the oxidizing agent (Cu^{2+}) in solution. However, if we physically separate the oxidizing agent from the reducing agent, the transfer of electrons can take place via an external conducting medium (a metal wire). As the reaction progresses, it sets up a constant flow of electrons and hence generates electricity (that is, it produces electrical work such as driving an electric motor).

The experimental apparatus for generating electricity through the use of a spontaneous reaction is called a **galvanic cell** or *voltaic cell*, after the Italian scientists Luigi Galvani and Alessandro Volta, who constructed early versions of the device. Figure 18.1 shows the essential components of a galvanic cell. A zinc bar is immersed in a ZnSO_4 solution, and a copper bar is immersed in a CuSO_4 solution. The cell operates on the principle that the oxidation of Zn to Zn^{2+} and the reduction of Cu^{2+} to Cu can be made to take place simultaneously in separate locations with the transfer of electrons between them occurring through an external wire. The zinc and copper bars are called *electrodes*. This particular arrangement of electrodes (Zn and Cu) and solutions (ZnSO_4 and CuSO_4) is called the Daniell cell. By definition, the **anode** in a galvanic cell is *the electrode at which oxidation occurs* and the **cathode** is *the electrode at which reduction occurs*.

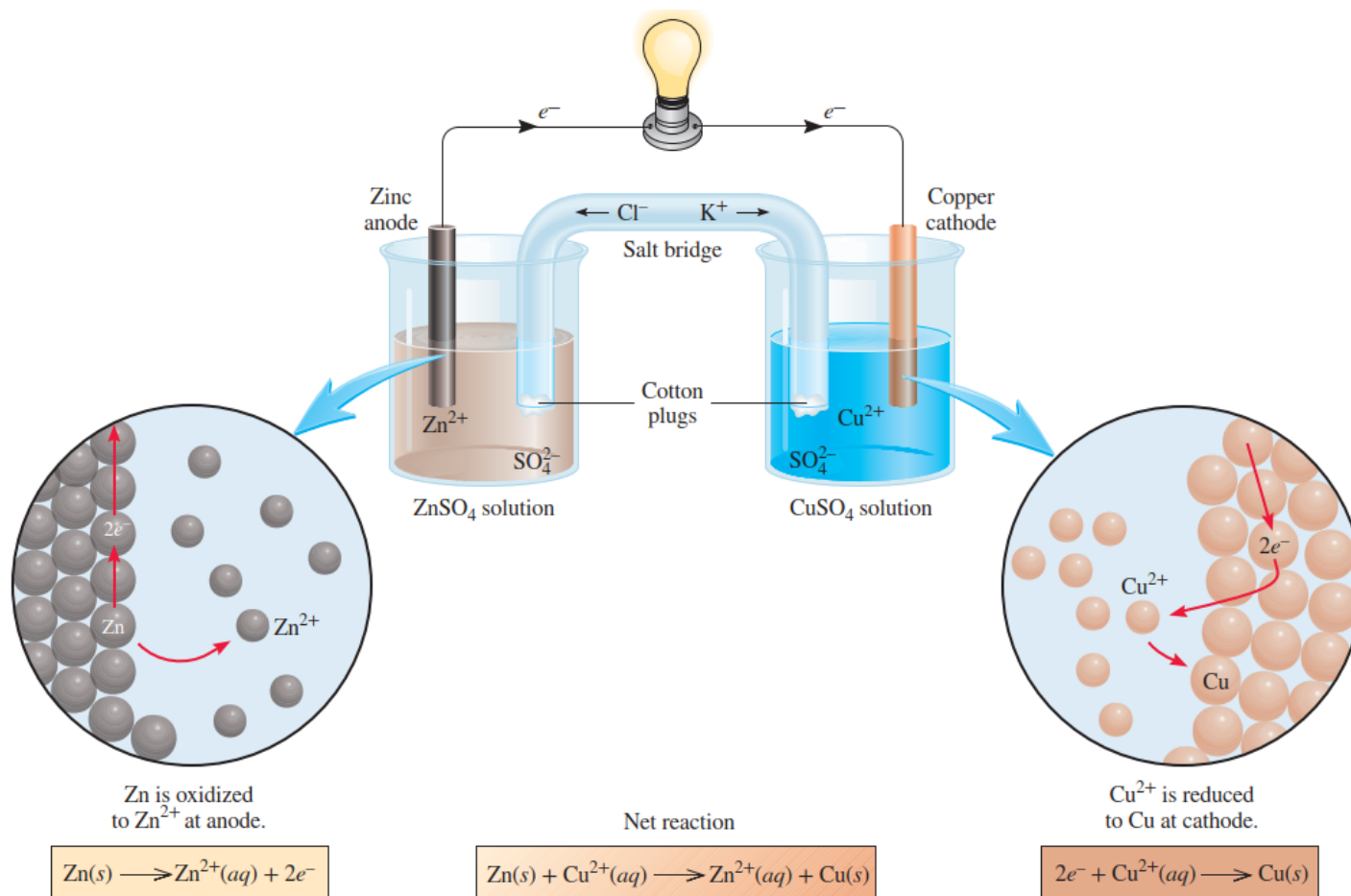
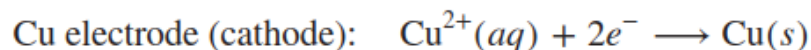


Figure 18.1 A galvanic cell. The salt bridge (an inverted U tube) containing a KCl solution provides an electrically conducting medium between two solutions. The openings of the U tube are loosely plugged with cotton balls to prevent the KCl solution from flowing into the containers while allowing the anions and cations to move across. The lightbulb is lit as electrons flow externally from the Zn electrode (anode) to the Cu electrode (cathode).

For the Daniell cell, the *half-cell reactions*, that is, *the oxidation and reduction reactions at the electrodes*, are

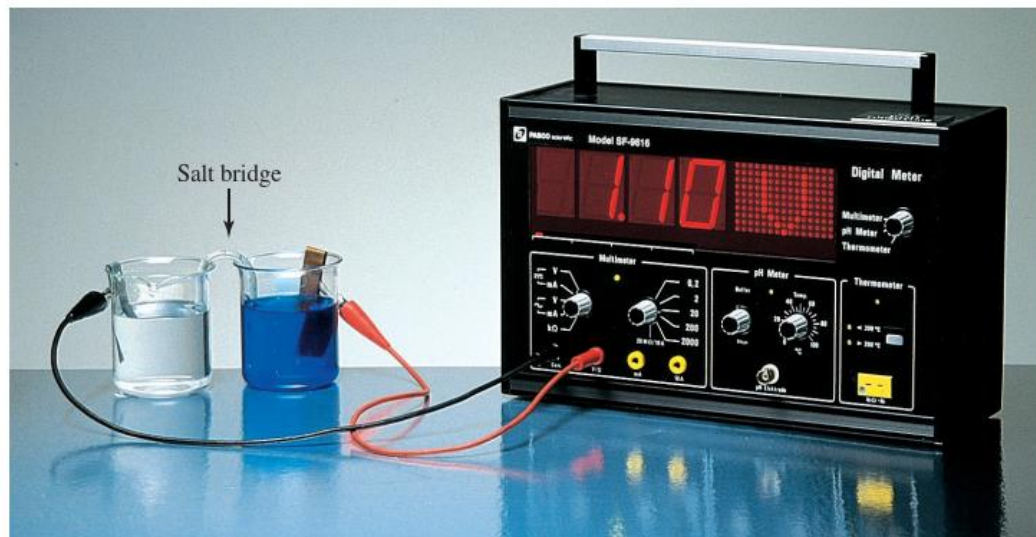


Note that unless the two solutions are separated from each other, the Cu^{2+} ions will react directly with the zinc bar:



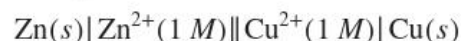
and no useful electrical work will be obtained.

To complete the electrical circuit, the solutions must be connected by a conducting medium through which the cations and anions can move from one electrode compartment to the other. This requirement is satisfied by a *salt bridge*, which, in its simplest form, is an inverted U tube containing an inert electrolyte solution, such as KCl or NH_4NO_3 , whose ions will not react with other ions in solution or with the electrodes (see Figure 18.1). During the course of the overall redox reaction, electrons flow externally from the anode (Zn electrode) through the wire to the cathode (Cu electrode). In the solution, the cations (Zn^{2+} , Cu^{2+} , and K^{+}) move toward the cathode, while the anions (SO_4^{2-} and Cl^{-}) move toward the anode. Without the salt bridge connecting the two solutions, the buildup of positive charge in the anode compartment (due to the formation of Zn^{2+} ions) and negative charge in the cathode compartment (created when some of the Cu^{2+} ions are reduced to Cu) would quickly prevent the cell from operating.



An electric current flows from the anode to the cathode because there is a difference in electrical potential energy between the electrodes. This flow of electric current is analogous to that of water down a waterfall, which occurs because there is a difference in gravitational potential energy, or the flow of gas from a high-pressure region to a low-pressure region. Experimentally, the *difference in electrical potential between the anode and the cathode* is measured by a voltmeter (Figure 18.2). The voltage across the electrodes of a galvanic cell is called the **cell voltage**, or **cell potential**. Another common term for cell voltage is the **electromotive force or emf (E)**, which, despite its name, is a measure of voltage, not force. We will see that the voltage of a cell depends not only on the nature of the electrodes and the ions, but also on the concentrations of the ions and the temperature at which the cell is operated.

The conventional notation for representing galvanic cells is the *cell diagram*. For the Daniell cell shown in Figure 18.1, if we assume that the concentrations of Zn^{2+} and Cu^{2+} ions are 1 *M*, the cell diagram is



The single vertical line represents a phase boundary. For example, the zinc electrode is a solid and the Zn^{2+} ions (from ZnSO_4) are in solution. Thus, we draw a line between Zn and Zn^{2+} to show the phase boundary. The double vertical lines denote the salt bridge. By convention, the anode is written first, to the left of the double lines, and the other components appear in the order in which we would encounter them in moving from the anode to the cathode.

▶ Voltaic Cells

The next several sections describe battery cells, or voltaic cells (also called galvanic cells). These are a kind of electrochemical cell. An **electrochemical cell** is a system consisting of electrodes that dip into an electrolyte and in which a chemical reaction either uses or generates an electric current. A **voltaic**, or **galvanic**, cell is an electrochemical cell in which a spontaneous reaction generates an electric current. An **electrolytic cell** is an electrochemical cell in which an external energy source drives an otherwise nonspontaneous reaction. In the next sections, we will discuss the basic principles behind voltaic cells and then explore some of their commercial uses.

19.3 Notation for Voltaic Cells

It is convenient to have a shorthand way of designating particular voltaic cells. The cell described earlier, consisting of a zinc metal–zinc ion half-cell and a copper metal–copper ion half-cell, is written



In this notation, the anode, or oxidation half-cell, is always written on the left; the cathode, or reduction half-cell, is written on the right. The two electrodes are electrically connected by means of a salt bridge, denoted by two vertical bars.



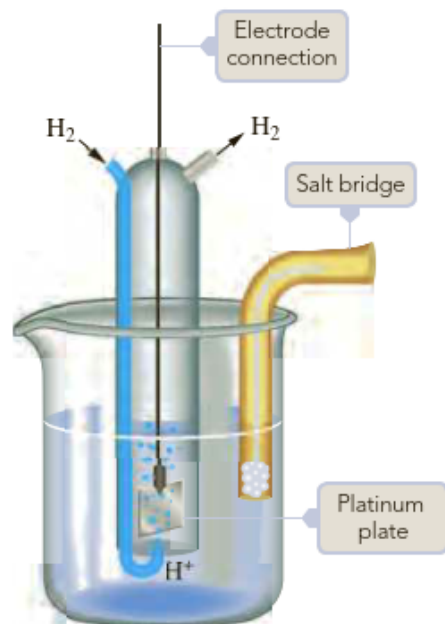


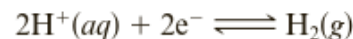
Figure 19.5 ▲

A hydrogen electrode Hydrogen gas bubbles over a platinum surface, where the half-reaction $2\text{H}^+(aq) + 2e^- \rightleftharpoons \text{H}_2(g)$ occurs.

The cell terminals are at the extreme ends in this cell notation, and a single vertical bar indicates a phase boundary—say, between a solid terminal and the electrode solution. For the anode of the same cell, you have



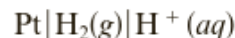
When the half-reaction involves a gas, an inert material such as platinum serves as a terminal and as an electrode surface on which the half-reaction occurs. The platinum catalyzes the half-reaction but otherwise is not involved in it. Figure 19.5 shows a hydrogen electrode; hydrogen bubbles over a platinum plate that is immersed in an acidic solution. The cathode half-reaction is



The notation for the hydrogen electrode, written as a cathode, is

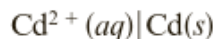


To write such an electrode as an anode, you simply reverse the notation:

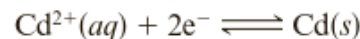
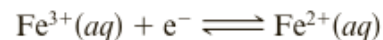
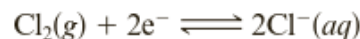


Here are several additional examples of this notation for electrodes (written as cathodes). A comma separates ions present in the same solution. We will write the cathode with oxidized species before reduced species, in the same order as in the half-reaction.

Cathode



Cathode Reaction



Example 19.4 Writing the Cell Reaction from the Cell Notation

Gaining Mastery Toolbox

Critical Concept 19.4

Cell notation fully describes the half-reactions of a voltaic cell. This notation can include the phases of the substances, ion concentrations, and gas pressures. The anode half-cell reaction is always represented on the left.

Solution Essentials:

- Cell notation
- Anode and cathode
- Salt bridge

a. Write the cell reaction for the voltaic cell



b. Write the cell reaction for the voltaic cell



Problem Strategy The cell notation gives the species involved in each half-reaction. Complete and balance each half-reaction, and then multiply by factors so that when you add the half-reactions, the electrons cancel. The result is the cell reaction.

(continued)

Example 19.4 (continued)

- Half-cell
- Voltaic cell
- Electrochemical cell
- Balanced oxidation–reduction reaction
- Half-reaction
- Oxidation–reduction reaction

Solution

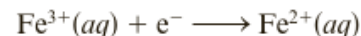
a. The half-cell reactions are



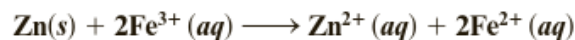
Multiplying the anode reaction by 2 and then summing the half-cell reactions gives



b. The half-cell reactions are



and the cell reaction is



19.4 Cell Potential

Work is needed to move electrons in a wire or to move ions through a solution to an electrode. The situation is analogous to pumping water from one point to another. Work must be expended to pump the water. The analogy may be extended. Water moves from a point at high pressure to a point at low pressure. Thus, a pressure difference is required. The work expended in moving water through a pipe depends on the volume of water and the pressure difference. The situation with electricity is similar. An electric charge moves from a point at high electric potential (high electrical pressure) to a point at low electric potential (low electrical pressure). The work needed to move an electric charge through a conductor depends on the total charge moved and the potential difference. **Potential difference** is *the difference in electric potential (electrical pressure) between two points*. You measure this quantity in volts. The **volt, V**, is *the SI unit of potential difference*. The electrical work expended in moving a charge through a conductor is

$$\text{Electrical work} = \text{charge} \times \text{potential difference}$$

Corresponding SI units for the terms in this equation are

$$\text{Joules} = \text{coulombs} \times \text{volts}$$

The **Faraday constant, F** , is *the magnitude of charge on one mole of electrons; it equals $9.6485 \times 10^4 \text{ C}$ per mole of electrons ($96,485 \text{ C/mol e}^-$)*. The *faraday* is a unit of charge equal to $9.6485 \times 10^4 \text{ C}$. In moving this quantity of charge (1 faraday of charge) from one electrode to another, the numerical value of the work done by a voltaic cell is the product of the faraday constant F times the potential



Figure 19.6 ▲

A digital voltmeter A digital voltmeter draws negligible current, so it can be used to measure cell potential.

difference between the electrodes. The work w is the negative of this, because the voltaic cell loses energy as it does work on the surroundings.

$$w = -F \times \text{potential difference}$$

In the normal operation of a voltaic cell, the potential difference (voltage) across the electrodes is less than the maximum possible voltage of the cell. One reason for this is that it takes energy or work to drive a current through the cell itself. The decrease in cell voltage as current is drawn reflects this energy expenditure within the cell; and the greater the current, the lower the voltage. Thus, the cell voltage has its maximum value only when no current flows. The situation is analogous to measuring the difference between the pressure of water in a faucet and that of the outside atmosphere. You expect this difference to be higher if you measure it by a device that does not require you to run water from the faucet. When you do run water, the pressure difference drops.

The maximum potential difference between the electrodes of a voltaic cell is referred to as the **cell potential** or **electromotive force (emf)** of the cell, or E_{cell} . It can be measured by an electronic digital voltmeter (Figure 19.6).

You can also determine which electrode of a cell is the anode and which is the cathode from this measurement. The digital voltmeter gives the reading with a sign attached, from which you can deduce the sign, or polarity, of each electrode. The anode of a voltaic cell has negative polarity, and the cathode has positive polarity.

We can now write an expression for the maximum work obtainable from a voltaic cell. Let n be the number of moles of electrons transferred in the overall cell equation. The maximum electrical work of a voltaic cell for molar amounts of reactants (according to the cell equation *as written*) is

$$w_{\text{max}} = -nFE_{\text{cell}}$$

Here E_{cell} is the cell potential, and F is the Faraday constant, $9.6485 \times 10^4 \text{ C/mol e}^-$.

18.4 Thermodynamics of Redox Reactions

Our next step is to see how E_{cell}° is related to thermodynamic quantities such as ΔG° and K . In a galvanic cell, chemical energy is converted to electrical energy to do electrical work such as running an electric motor. Electrical energy, in this case, is the product of the emf of the cell and the total electrical charge (in coulombs) that passes through the cell:

$$\begin{aligned}\text{electrical energy} &= \text{coulombs} \times \text{volts} \\ &= \text{joules}\end{aligned}$$

The equality means that

$$1 \text{ J} = 1 \text{ C} \times 1 \text{ V}$$

The total charge is determined by the number of electrons that pass through the cell, so we have

$$\text{total charge} = \text{number of } e^{-} \times \text{charge of one } e^{-}$$

In general, it is more convenient to express the total charge in molar quantities. The charge of one mole of electrons is called the **Faraday constant (F)**, after the English chemist and physicist Michael Faraday,[†] where

$$\begin{aligned}1 F &= 6.022 \times 10^{23} e^{-}/\text{mol } e^{-} \times 1.602 \times 10^{-19} \text{ C}/e^{-} \\ &= 9.647 \times 10^4 \text{ C/mol } e^{-}\end{aligned}$$

Therefore, the total charge can now be expressed as nF , where n is the number of moles of electrons exchanged between the oxidizing agent and reducing agent in the overall redox equation for the electrochemical process.

The measured emf (E_{cell}) is the *maximum* voltage the cell can achieve. Therefore, the electrical work done, w_{ele} , which is the maximum work that can be done (w_{max}), is given by the product of the total charge and the emf of the cell:

$$w_{\text{max}} = w_{\text{ele}} = -nFE_{\text{cell}}$$

The negative sign indicates that the electrical work is done by the system (galvanic cell) on the surroundings. In Chapter 17 we defined free energy as the energy available to do work. Specifically, the change in free energy (ΔG) represents the maximum amount of useful work that can be obtained in a reaction:

$$\Delta G = w_{\max} = w_{\text{ele}}$$

Therefore, we can write

$$\Delta G = -nFE_{\text{cell}} \quad (18.2)$$

For a spontaneous reaction, ΔG is negative. Because both n and F are positive quantities, it follows that E_{cell} must also be positive. For reactions in which reactants and products are in their standard states (1 M or 1 atm), Equation (18.2) becomes

$$\Delta G^{\circ} = -nFE_{\text{cell}}^{\circ} \quad (18.3)$$

Now we can relate E_{cell}° to the equilibrium constant (K) of a redox reaction. In Section 17.5 we saw that the standard free-energy change ΔG° for a reaction is related to its equilibrium constant as follows [see Equation (17.14)]:

$$\Delta G^{\circ} = -RT \ln K$$

If we combine Equations (17.14) and (18.3), we obtain

$$-nFE_{\text{cell}}^{\circ} = RT \ln K$$

Solving for E_{cell}°

$$E_{\text{cell}}^{\circ} = \frac{RT}{nF} \ln K \quad (18.4)$$

When $T = 298 \text{ K}$, Equation (18.4) can be simplified by substituting for R and F :

$$E_{\text{cell}}^{\circ} = \frac{(8.314 \text{ J/K} \cdot \text{mol})(298 \text{ K})}{n(96,500 \text{ J/V} \cdot \text{mol})} \ln K$$

or

$$E_{\text{cell}}^{\circ} = \frac{0.0257 \text{ V}}{n} \ln K \quad (18.5)$$

Alternatively, Equation (18.5) can be written using the base-10 logarithm of K :

$$E_{\text{cell}}^{\circ} = \frac{0.0592 \text{ V}}{n} \log K \quad (18.6)$$

Thus, if any one of the three quantities ΔG° , K , or E_{cell}° is known, the other two can be calculated using Equation (17.14), Equation (18.3), or Equation (18.4) (Figure 18.5). We summarize the relationships among ΔG° , K , and E_{cell}° and characterize the spontaneity of a redox reaction in Table 18.2. For simplicity, we sometimes omit the subscript “cell” in E° and E .

Table 18.2

Relationships Among ΔG° , K , and E°_{cell}

ΔG°	K	E°_{cell}	Reaction Under Standard-State Conditions
Negative	>1	Positive	Favors formation of products.
0	$=1$	0	Reactants and products are equally favored.
Positive	<1	Negative	Favors formation of reactants.

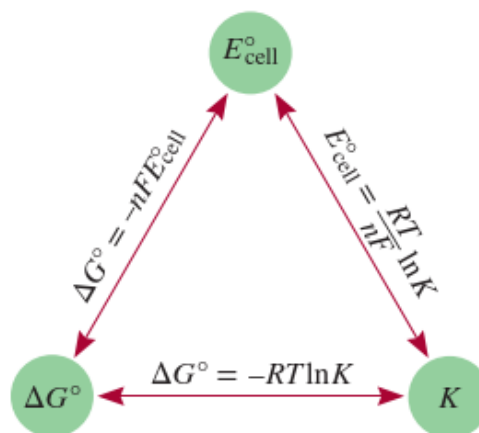
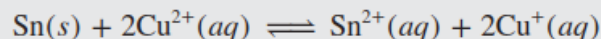


Figure 18.5 Relationships among E° , K , and ΔG° .

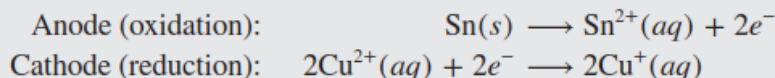
Example 18.4

Calculate the equilibrium constant for the following reaction at 25°C:



Strategy The relationship between the equilibrium constant K and the standard emf is given by Equation (18.5): $E_{\text{cell}}^{\circ} = (0.0257 \text{ V}/n)\ln K$. Thus, if we can determine the standard emf, we can calculate the equilibrium constant. We can determine the E_{cell}° of a hypothetical galvanic cell made up of two couples (Sn^{2+}/Sn and $\text{Cu}^{2+}/\text{Cu}^+$) from the standard reduction potentials in Table 18.1.

Solution The half-cell reactions are



$$\begin{aligned}E_{\text{cell}}^{\circ} &= E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ} \\ &= E_{\text{Cu}^{2+}/\text{Cu}^+}^{\circ} - E_{\text{Sn}^{2+}/\text{Sn}}^{\circ} \\ &= 0.15 \text{ V} - (-0.14 \text{ V}) \\ &= 0.29 \text{ V}\end{aligned}$$

Equation (18.5) can be written

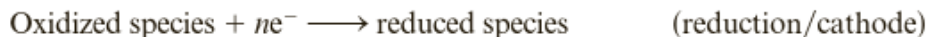
$$\ln K = \frac{nE^{\circ}}{0.0257 \text{ V}}$$

In the overall reaction we find $n = 2$. Therefore,

$$\begin{aligned}\ln K &= \frac{(2)(0.29 \text{ V})}{0.0257 \text{ V}} = 22.6 \\ K &= e^{22.6} = 7 \times 10^9\end{aligned}$$

19.5 Standard Cell Potentials and Standard Electrode Potentials

A cell potential is a measure of the driving force of the cell reaction. This reaction occurs in the cell as separate half-reactions: an oxidation half-reaction and a reduction half-reaction. The general forms of these half-reactions are



You can imagine that the cell potential is composed of a contribution from the anode (whose value depends on the ability of the oxidation half-reaction to lose electrons) and a contribution from the cathode (whose value depends on the ability of the reduction half-reaction to gain electrons). We call these contributions the *oxidation potential* and the *reduction potential*, respectively. Then

$$E_{\text{cell}} = \text{oxidation potential} + \text{reduction potential}$$

A reduction potential is a measure of the tendency for an oxidized species to gain electrons in the reduction half-reaction (in other words, a measure of the ability of the species to act as an oxidizing agent). If you can construct a table of reduction potentials, you will have a list of strengths of oxidizing agents, in addition to having a way of calculating cell potentials.

You can look at an oxidation half-reaction as the reverse of a corresponding reduction half-reaction. The oxidation potential for an oxidation half-reaction equals the negative of the reduction potential for the reverse half-reaction (which is a reduction).

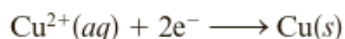
$$\text{Oxidation potential for a half-reaction} = -\text{reduction potential for the reverse half-reaction}$$

This means that in practice you need to tabulate only oxidation potentials or reduction potentials. The choice, by convention, is to tabulate reduction potentials. We call these *electrode potentials*, and we denote them by the symbol E (without the subscript cell, as in E_{cell}).

Consider the zinc–copper cell described earlier.



The half-reactions are



The first half-reaction is an oxidation. If you write E_{Zn} for the electrode potential corresponding to the reduction half-reaction $\text{Zn}^{2+}(aq) + 2e^{-} \longrightarrow \text{Zn}(s)$, then $-E_{\text{Zn}}$ is the potential for the oxidation half-reaction $\text{Zn}(s) \longrightarrow \text{Zn}^{2+}(aq) + 2e^{-}$. The copper half-reaction is a reduction. Write E_{Cu} for the electrode potential.

The cell potential is the sum of the potentials for the reduction and oxidation half-reactions. For the cell we have been describing, the cell potential is the sum of the reduction potential (electrode potential) for the copper half-cell and the oxidation potential (negative of the electrode potential) for the zinc half-cell.

$$E_{\text{cell}} = E_{\text{Cu}} + (-E_{\text{Zn}}) = E_{\text{Cu}} - E_{\text{Zn}}$$

Note that the cell potential equals the difference between the two electrode potentials. You can think of the electrode potential as the electric potential on the electrode, and you obtain the cell potential as a potential difference in which you subtract the anode potential from the cathode potential.

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}}$$

The electrode potential is an *intensive property*. This means that its value is independent of the amount of species in the reaction. Thus, the electrode potential for the half-reaction



is the same as for



Tabulating Standard Electrode Potentials

The cell potential of a voltaic cell depends on the concentrations of substances and the temperature of the cell. For purposes of tabulating electrochemical data, it is usual to choose thermodynamic standard-state conditions for voltaic cells. The **standard cell potential**, E°_{cell} , is the *emf of a voltaic cell operating under standard-state conditions (solute concentrations are each 1 M, gas pressures are each 1 atm, and the temperature has a specified value—usually 25°C)*. Note the superscript degree sign ($^\circ$), which signifies standard-state conditions. ◀

If you can derive a table of electrode potentials, you can calculate cell potentials from them. This provides a great advantage over tabulating cell potentials. From a small table of electrode potentials, you can obtain all the cell potentials you could construct from pairs of electrodes. For instance, a table of 40 electrode potentials would give the cell potentials of nearly 800 voltaic cells.

However, it is not possible to measure the potential of a single electrode; only the cell potentials of cells can be measured. What you can do is measure the cell potentials of cells constructed from various electrodes connected in turn to one particular electrode, which you choose as a reference. You arbitrarily assign this reference electrode a potential equal to zero and obtain the potentials for the other electrodes by measuring the cell potentials. By convention, the reference chosen for comparing electrode potentials is the *standard hydrogen electrode*, and it is assigned a potential of 0.00 V. This is an electrode like that pictured earlier in Figure 19.5 operating under standard-state conditions.

The **standard electrode potential**, E° , is the *electrode potential when the concentrations of solutes are 1 M, the gas pressures are 1 atm, and the temperature has a*

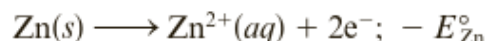
specified value (usually 25°C). The superscript degree sign ($^{\circ}$) signifies standard-state conditions.

To understand how standard electrode potentials are obtained, look at how you would find the standard electrode potential, E° , for the zinc electrode. You connect a standard zinc electrode to a standard hydrogen electrode. When you measure the cell potential with a voltmeter, you obtain 0.76 V, with the zinc electrode acting as the anode.

Now write the cell potential in terms of the electrode potentials. The cell is



and the half-reactions with corresponding half-cell potentials (oxidation or reduction potentials) are



The cell potential is the sum of the half-cell potentials.

$$E_{\text{cell}} = E_{\text{H}_2}^{\circ} + (-E_{\text{Zn}}^{\circ})$$

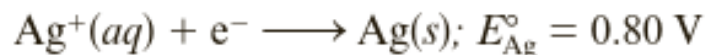
Substitute 0.76 V for the cell potential and 0.00 V for the standard hydrogen electrode potential. This gives $E_{\text{Zn}}^{\circ} = -0.76$ V.

Proceeding in this way, you can obtain the electrode potential for a series of half-cell reactions. Table 19.1 lists standard electrode potentials for selected half-cells at 25°C.

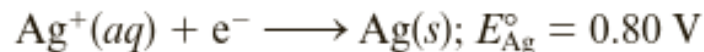
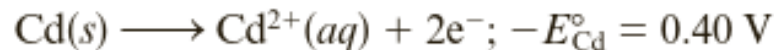
Calculating Cell Potentials by Addition of Half-Reactions and Standard Potentials Consider a voltaic cell constructed from a cadmium electrode and a silver electrode having the following reduction half-reactions and corresponding standard electrode potentials (reduction potentials):



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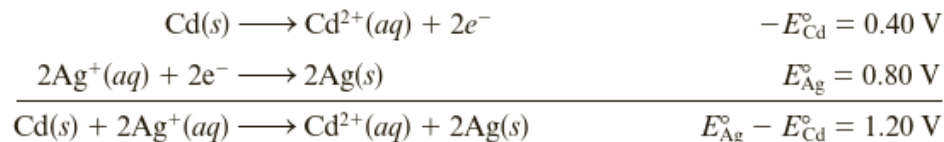


You will need to reverse one of these half-reactions to obtain the oxidation part of the cell reaction. The cell reaction is spontaneous, with the stronger reducing agent on the left. This will be Cd, because it is the reactant in the half-reaction with the more negative electrode potential. Therefore, you reverse the first half-reaction, as well as the sign of the half-cell potential (to obtain the oxidation potential, which equals $-E_{\text{Cd}}^{\circ}$).



To obtain the cell reaction, you must multiply the half-reactions by factors so that when the half-reactions are added together, the electrons cancel. This does not

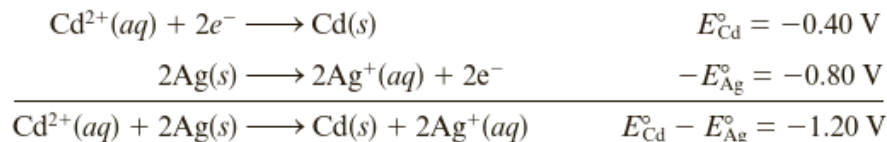
affect the half-cell potentials, which do not depend on amount of substance. You multiply the second half-reaction by 2 and then add.



The cell potential, E_{cell}° , is 1.20 V. The cell diagram is



Suppose that at the beginning of this problem, you had mistakenly reversed the silver half-reaction, instead of correctly reversing the cadmium half-reaction. Then your work would have looked like the following:



The corresponding cell notation would have been



Everything is simply the reverse of the correct result. At this point, you realize your mistake, because the cell potential is found to be negative. A negative potential merely indicates that the cell reaction is nonspontaneous as written. To obtain the spontaneous reaction and a positive cell potential, you simply reverse both half-reactions and corresponding half-cell potentials. (You would also reverse the cell notation.) This changes the sign of the cell potential.

Calculation of Cell Potentials Using Standard Reduction Potentials Note that in the case of the cadmium–silver cell above, the cell potential equals the standard electrode potential of the cathode minus the standard electrode potential of the anode. Thus, alternatively, you can calculate any cell potential using the equation

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$$

Applying this equation to our example, you obtain

$$E_{\text{cell}}^{\circ} = E_{\text{Ag}}^{\circ} - E_{\text{Cd}}^{\circ} = 0.80 \text{ V} - (-0.40 \text{ V}) = 1.20 \text{ V}$$

Example 19.8 Calculating the Cell Potential from Standard Potentials

Gaining Mastery Toolbox

Critical Concept 19.8

A voltaic cell always has a positive standard cell potential. When employing the addition of half-reaction approach, the half-reactions must be combined in such a way that the sum of the half-cell potentials add up to a voltage greater than zero. Direct calculation of the standard cell potential can be accomplished by using the equation

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$$

Solution Essentials:

- Standard cell potential (E_{cell}°)
- Direct calculation of standard cell potential ($E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$)
- Table of Standard Electrode (Reduction) Potentials (Table 19.1)

Calculate the standard cell potential of the following voltaic cell at 25°C using standard electrode potentials.



What is the cell reaction?

Problem Strategy From a table of electrode potentials, write the two reduction half-reactions and standard electrode potentials for the cell. The cell notation assumes that the anode (the oxidation half-cell) is on the left. Change the direction of this half-reaction and the sign of its electrode potential. (Assuming that the cell notation was written correctly, you change the direction of the half-reaction corresponding to the smaller, or more negative, electrode potential.) Multiply the half-reactions (but not the electrode potentials) by factors so that when the half-reactions are added, the electrons cancel. The sum of the half-reactions is the cell reaction. Add the electrode potentials to get the cell potential.

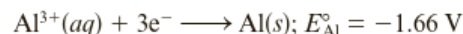
An alternative strategy is to apply the equation

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$$

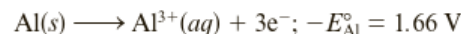
(continued)

- Cell notation
- Voltaic cell
- Anode and cathode
- Balanced oxidation–reduction reaction
- Half-reaction

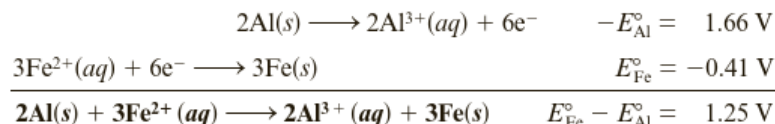
Solution (Both Approaches) The reduction half-reactions and standard electrode potentials are



You reverse the first half-reaction and its half-cell potential to obtain



You obtain the cell potential by adding the half-cell potentials. Because you also want the cell reaction, you multiply the first half-reaction by 2 and the second half-reaction by 3, so that when the half-reactions are added, the electrons cancel. The addition of half-reactions is displayed below. The cell potential is **1.25 V**.



Alternatively, you can calculate the cell potential from the formula

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$$

to obtain

$$\begin{aligned} E_{\text{cell}}^{\circ} &= E_{\text{Fe}}^{\circ} - E_{\text{Al}}^{\circ} \\ &= -0.41 \text{ V} - (-1.66 \text{ V}) \\ &= 1.25 \text{ V} \end{aligned}$$

18.6 Batteries and Fuel Cells

A **battery** is a galvanic cell, or a series of combined galvanic cells, that can be used as a source of direct electric current at a constant voltage. Although the operation of a battery is similar in principle to that of the galvanic cells described in Section 18.2, a battery has the advantage of being completely self-contained and requiring no auxiliary components such as salt bridges. Here we will discuss several types of batteries that are in widespread use.

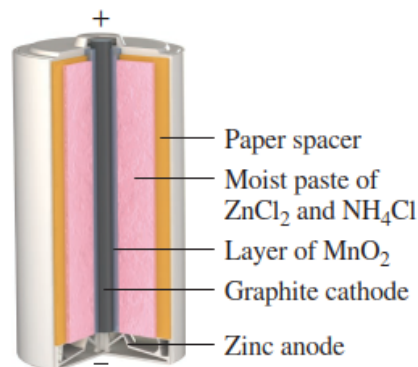
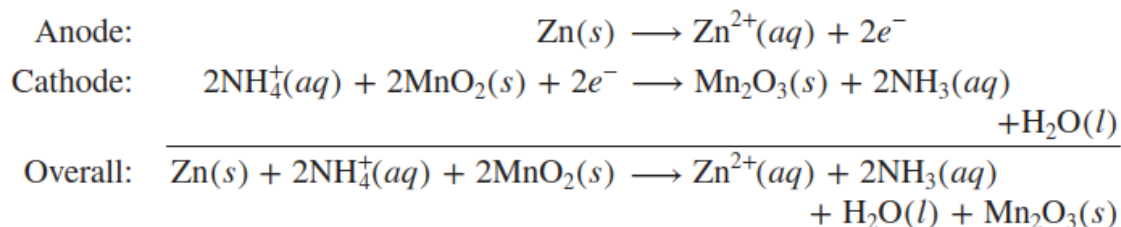


Figure 18.7 Interior section of a dry cell of the kind used in flashlights and transistor radios. Actually, the cell is not completely dry, as it contains a moist electrolyte paste.

The Dry Cell Battery

The most common dry cell, that is, a cell without a fluid component, is the *Leclanché cell* used in flashlights and transistor radios. The anode of the cell consists of a zinc can or container that is in contact with manganese dioxide (MnO₂) and an electrolyte. The electrolyte consists of ammonium chloride and zinc chloride in water, to which starch is added to thicken the solution to a pastelike consistency so that it is less likely to leak (Figure 18.7). A carbon rod serves as the cathode, which is immersed in the electrolyte in the center of the cell. The cell reactions are



Actually, this equation is an oversimplification of a complex process. The voltage produced by a dry cell is about 1.5 V.

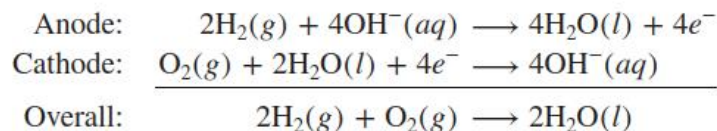
Fuel Cells

Fossil fuels are a major source of energy, but conversion of fossil fuel into electrical energy is a highly inefficient process. Consider the combustion of methane:



To generate electricity, heat produced by the reaction is first used to convert water to steam, which then drives a turbine that drives a generator. An appreciable fraction of the energy released in the form of heat is lost to the surroundings at each step; even the most efficient power plant converts only about 40 percent of the original chemical energy into electricity. Because combustion reactions are redox reactions, it is more desirable to carry them out directly by electrochemical means, thereby greatly increasing the efficiency of power production. This objective can be accomplished by a device known as a ***fuel cell***, a *galvanic cell that requires a continuous supply of reactants to keep functioning*.

In its simplest form, a hydrogen-oxygen fuel cell consists of an electrolyte solution, such as potassium hydroxide solution, and two inert electrodes. Hydrogen and oxygen gases are bubbled through the anode and cathode compartments (Figure 18.11), where the following reactions take place:



The standard emf of the cell can be calculated as follows, with data from Table 18.1:

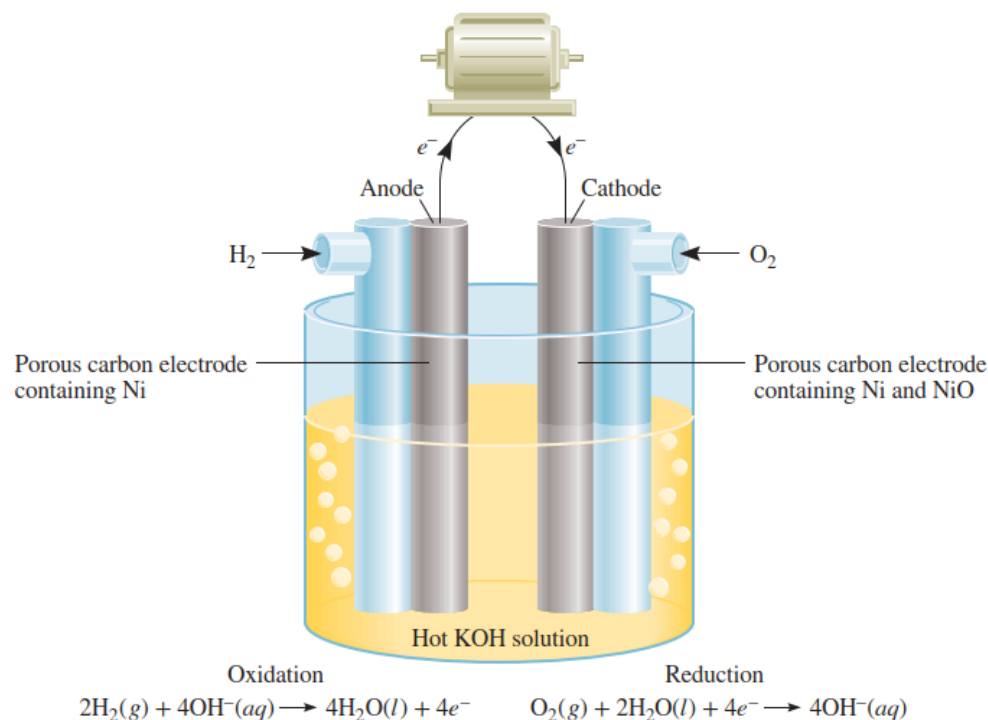
$$\begin{aligned} E^\circ_{\text{cell}} &= E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} \\ &= 0.40 \text{ V} - (-0.83 \text{ V}) \\ &= 1.23 \text{ V} \end{aligned}$$



A car powered by hydrogen fuel cells manufactured by General Motors.

©Michael Klinec/Alamy Stock Photo

Figure 18.11 A hydrogen-oxygen fuel cell. The Ni and NiO embedded in the porous carbon electrodes are electrocatalysts.



Thus, the cell reaction is spontaneous under standard-state conditions. Note that the reaction is the same as the hydrogen combustion reaction, but the oxidation and reduction are carried out separately at the anode and the cathode. Like platinum in the standard hydrogen electrode, the electrodes have a twofold function. They serve as electrical conductors, and they provide the necessary surfaces for the initial decomposition of the molecules into atomic species, prior to electron transfer. They are *electrocatalysts*. Metals such as platinum, nickel, and rhodium are good electrocatalysts.